Determinants

A determinants is a number Calculated from a a square matrix that give important information about the matrix

Symbol using vertical bars:- $|A|$

Example

2D Matrix In 2D matrix simply Cross mutiply

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}_{2\times 2} \qquad \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

$|A| = (2\times 5) - (3\times 4)$ (cross mutiplication

$|A| = 10 - 12$

$|A| = -2$

3D

$$A = \begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{bmatrix}$$

Here In 1 step 8 row →

row and column ↳ leave write rest ↓ all same as column

$-8 \rightarrow$, $-2 \rightarrow$

Cross mutiply 3X3

step

1 $|A| = 8\begin{vmatrix} -3 & -2 \\ -4 & 1 \end{vmatrix} - (-8)\begin{vmatrix} 4 & -2 \\ 3 & 1 \end{vmatrix} + (-2)\begin{vmatrix} 4 & -3 \\ 3 & -4 \end{vmatrix}$

2 $|A| = 8(-3-8) + 8(4+6) - 2(-16+9)$

3 $|A| = 6$

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$|A| = a(ei - fg) - b(di - fg) + c(dh - eg)$$

where the the + − + Comes from from:

Sign pattern for determinants

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Most of time we use the first row + − +, it is easier to understand and standard.